Worksheet 6.4

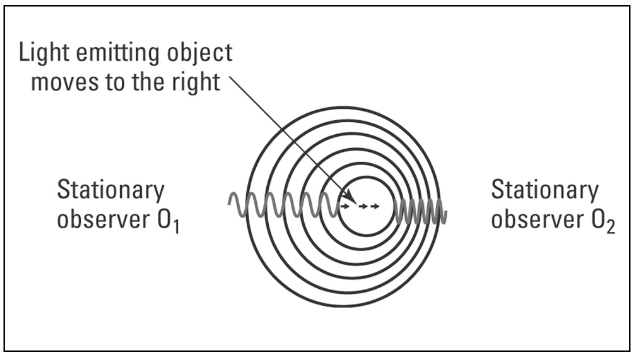
The expanding universe

Student: .... Class: ....

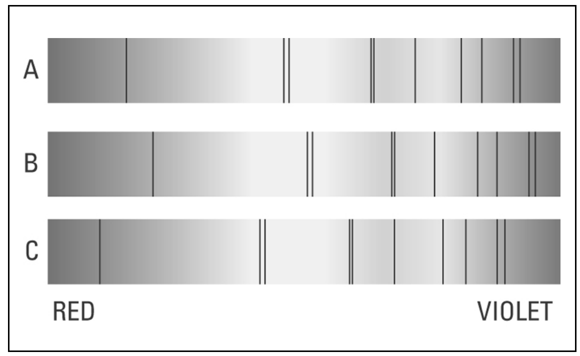
The Doppler effect

1. Figure A shows a light emitting object moving to the right. The light emitted is detected by two stationary observers (O_1 and O_2). Figure B shows absorption spectra of a light emitting object. The dark lines represent light colours that have been absorbed by different elements in the star.

A



B



(a) Explain the Doppler effect using figure A.

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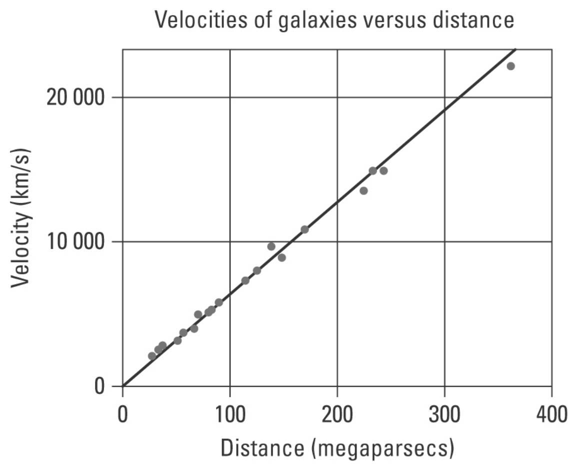
(b) The dark lines in the spectra in figure B show red and blue spectral shifting. One spectrum is of a light source that is stationary relative to the observer. Identify the spectrum (X, Y or Z) in which the light emitting source is moving:

(i) away from the observer ....

(ii) towards the observer. ....

Hubble’s expanding universe

2. The following graph shows the velocities of a number of galaxies moving away from our galaxy versus their distance from the Earth.



(a) Describe the trend demonstrated by the graph.

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(b) Explain why these galaxies are moving away from our own.

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